

Supplementary material

DriftJax: A Structure-Aware Differentiable Drift–Diffusion Solver for Photovoltaic Device Design

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This supplement accompanies both the standard and Elsevier CAS versions of the article. Sections S1–S6 develop the mathematical conventions and verification procedures; S7 presents additional device examples; S8 records the provenance of retained numerical values; and S9 specifies the information required for reproduction. Derivations are distinguished from implementation claims and from reported measurements. Implementation descriptions follow the supplied v0.1.15 release documents; the source tree and original run logs were not available for an independent executable audit. No additional simulation data are introduced.

S1 Physical conventions and nondimensionalisation

S1.1 State variables and units

The device occupies $0 \leq x \leq d$. Electron and hole electrical current densities are positive along $+x$, and the elementary charge q is positive. In these conventions, the standard transport equations are [1, 2]

$$\partial_x(\varepsilon \partial_x \phi) = q(n - p + N_A^- - N_D^+), \quad (\text{S1.1})$$

$$\partial_x J_n = q(R - G), \quad \partial_x J_p = q(G - R). \quad (\text{S1.2})$$

The electrostatic and quasi-Fermi potentials are distinct variables. For the energy convention $E_c = -\chi - q\phi$, $E_v = E_c - E_g$, $E_{Fn} = q\phi_n$, and $E_{Fp} = q\phi_p$, the non-degenerate densities are [1, 2]

$$n = N_c \exp\left(\frac{q\phi_n + \chi + q\phi}{k_B T}\right), \quad p = N_v \exp\left(\frac{-q\phi_p - \chi - E_g - q\phi}{k_B T}\right). \quad (\text{S1.3})$$

An alternative sign convention for quasi-Fermi potentials is possible, but the density formulas, current definitions, and boundary conditions must then be transformed together. For a homogeneous Boltzmann material, Eq. (S1.3) gives $J_n = q\mu_n n \partial_x \phi_n$ and $J_p = q\mu_p p \partial_x \phi_p$.

S1.2 Reference scales

Choose a density n_0 , reference permittivity ε_{ref} , and mobility μ_0 . A consistent scaling is

$$V_T = \frac{k_B T}{q}, \quad L_0 = \sqrt{\frac{\varepsilon_{\text{ref}} k_B T}{q^2 n_0}}, \quad (\text{S1.4})$$

$$t_0 = \frac{L_0^2}{\mu_0 V_T}, \quad J_0 = \frac{q\mu_0 n_0 V_T}{L_0}, \quad R_0 = \frac{n_0}{t_0}. \quad (\text{S1.5})$$

Table S1: Principal dimensional quantities. SI units apply to the equations; displayed semiconductor units require explicit conversion.

Symbol	Meaning	SI unit
x, d	Position and device thickness	m
ϕ, ϕ_n, ϕ_p, V	Potentials and applied voltage	V
$n, p, N_c, N_v, N_D^+, N_A^-$	Carrier, state, and dopant densities	m ⁻³
ε	Absolute permittivity	F m ⁻¹
χ, E_g, E_t	Electron affinity, band gap, trap energy	J
μ_n, μ_p	Mobilities	m ² V ⁻¹ s ⁻¹
J_n, J_p	Electrical current densities	A m ⁻²
G, R	Generation and recombination rates	m ⁻³ s ⁻¹
S_n, S_p	Contact exchange velocities	m s ⁻¹
I_λ	Spectral irradiance per unit wavelength	W m ⁻³

Let $\hat{x} = x/L_0$, $\hat{t} = t/t_0$, $\hat{\phi} = \phi/V_T$, $\hat{n} = n/n_0$, $\hat{\mu} = \mu/\mu_0$, and $\hat{\varepsilon} = \varepsilon/\varepsilon_{\text{ref}}$. The other potentials and carrier densities use the corresponding scales. Then

$$\partial_{\hat{x}}(\hat{\varepsilon}\partial_{\hat{x}}\hat{\phi}) = \hat{n} - \hat{p} + \hat{N}_A^- - \hat{N}_D^+, \quad (\text{S1.6})$$

$$\partial_{\hat{x}}\hat{J}_n = \hat{R} - \hat{G}, \quad \partial_{\hat{x}}\hat{J}_p = \hat{G} - \hat{R}. \quad (\text{S1.7})$$

Energy parameters in electronvolts are converted consistently before forming ratios with $k_B T$; one must not divide an electronvolt value by an energy in joules without conversion. For the illustrative choice $\varepsilon_{\text{ref}} = \varepsilon_0$, $n_0 = 10^{19} \text{ cm}^{-3}$, and $T = 300 \text{ K}$, this formula gives $L_0 \simeq 0.378 \text{ nm}$. This arithmetic check is not a claim that the unavailable implementation uses that particular length scale.

S1.3 Carrier statistics and their derivatives

Define the normalised complete Fermi–Dirac integral [3] by

$$F_s(\eta) = \frac{1}{\Gamma(s+1)} \int_0^\infty \frac{t^s}{1 + \exp(t-\eta)} dt, \quad s > -1. \quad (\text{S1.8})$$

For parabolic bands [4], $n = N_c F_{1/2}(\eta_n)$ and $p = N_v F_{1/2}(\eta_p)$ with $\eta_n = (E_{Fn} - E_c)/(k_B T)$ and $\eta_p = (E_v - E_{Fp})/(k_B T)$. This convention gives

$$\frac{dF_{1/2}}{d\eta} = F_{-1/2}(\eta), \quad F_{1/2}(\eta) \sim e^\eta \quad (\eta \rightarrow -\infty). \quad (\text{S1.9})$$

There is no second normalisation factor in the density formula. The documented numerical implementation uses quadrature and an asymptotic branch; its value and derivative should be checked over the actual degeneracy range, including the branch transition.

For Boltzmann statistics, the derivatives at fixed material parameters are especially simple:

$$\frac{\partial \hat{n}}{\partial(\hat{\phi}_n, \hat{\phi}_p, \hat{\phi})} = (\hat{n}, 0, \hat{n}), \quad \frac{\partial \hat{p}}{\partial(\hat{\phi}_n, \hat{\phi}_p, \hat{\phi})} = (0, -\hat{p}, -\hat{p}). \quad (\text{S1.10})$$

For numerical Fermi–Dirac statistics, replace these density factors by $\hat{N}_c F_{-1/2}(\eta_n)$ and $\hat{N}_v F_{-1/2}(\eta_p)$ with the same signs. The legacy Blakemore closure requires derivatives of its actual implemented approximation. Non-degenerate Einstein and mass-action formulas are not automatically valid after changing only the density closure [4].

S1.4 Recombination and equilibrium

The non-degenerate SRH trap model [5, 6], together with the radiative and Auger terms used in conventional device models [1], gives

$$R_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}, \quad (\text{S1.11})$$

$$R_{\text{rad}} = B_r(np - n_i^2), \quad (\text{S1.12})$$

$$R_{\text{Auger}} = (C_n n + C_p p)(np - n_i^2), \quad (\text{S1.13})$$

where $n_1 = n_i \exp[(E_t - E_i)/(k_B T)]$ and $p_1 = n_i \exp[-(E_t - E_i)/(k_B T)]$. At Boltzmann thermal equilibrium, $np = n_i^2$ and these net rates vanish. Under degenerate statistics, equilibrium consistency requires a compatible recombination closure rather than an unqualified use of this identity.

For analytical assembly, set $A = np - n_i^2$ and $D = \tau_p(n + n_1) + \tau_n(p + p_1)$. At fixed material parameters,

$$\partial_n R_{\text{SRH}} = (pD - A\tau_p)/D^2, \quad \partial_p R_{\text{SRH}} = (nD - A\tau_n)/D^2, \quad (\text{S1.14})$$

$$\partial_n R_{\text{rad}} = B_r p, \quad \partial_p R_{\text{rad}} = B_r n, \quad (\text{S1.15})$$

$$\partial_n R_{\text{Auger}} = C_n A + (C_n n + C_p p)p, \quad \partial_p R_{\text{Auger}} = C_p A + (C_n n + C_p p)n. \quad (\text{S1.16})$$

State derivatives follow by the chain rule through the selected density model. Parameter derivatives must additionally include n_i , trap occupancies, coefficients, and any parameter-dependent scaling.

S1.5 Generation and contacts

For spectral irradiance I_λ , the single-pass Beer–Lambert generation model without front reflection is [2]

$$G(x) = \int \frac{\lambda I_\lambda}{hc} \alpha(\lambda, x) \exp \left[- \int_0^x \alpha(\lambda, x') dx' \right] d\lambda. \quad (\text{S1.17})$$

A numerical sum must include wavelength weights. The incident power is $P_{\text{in}} = \int I_\lambda d\lambda$ over the declared incident spectrum. If generation is integrated over a truncated wavelength interval, the choice of denominator for efficiency must still be stated. Optical interpolation, smoothed absorption onsets, and thickness-dependent attenuation all contribute to $\mathbf{F}_\theta$. Coherent transfer matrices require complex refractive indices and consistent energy-flux normalisation [7].

Let $\nu = -1$ at the left boundary and $\nu = +1$ at the right. A dimensional outward carrier-exchange convention for the usual surface-recombination boundary model [1, 2] is

$$J_n \nu = -q S_n (n - n_{\text{eq}}), \quad J_p \nu = q S_p (p - p_{\text{eq}}). \quad (\text{S1.18})$$

The opposite signs reflect electron and hole charge. Ohmic reservoirs fix quasi-Fermi levels and equilibrium carrier populations; charge neutrality determines an associated electrostatic boundary value. Work-function offsets modify that value for the documented Schottky option. Equating the quasi-Fermi and electrostatic potentials is not a general Ohmic boundary condition. The four release parameters **Snl**, **Snr**, **Sp1**, and **Spr** must be mapped to the same current and normal conventions used in the interior residual.

S2 Discretisation and analytical Jacobian

S2.1 Edge fluxes

For a homogeneous Boltzmann segment with spacing h_i and $\Delta_i = (\phi_{i+1} - \phi_i)/V_T$, the Scharfetter–Gummel fitted currents are [8]

$$J_{n,i+1/2} = a_{n,i} [B(\Delta_i)n_{i+1} - B(-\Delta_i)n_i], \quad (\text{S2.1})$$

$$J_{p,i+1/2} = a_{p,i} [B(\Delta_i)p_i - B(-\Delta_i)p_{i+1}], \quad (\text{S2.2})$$

where $a_{n,p,i} = q\mu_{n,p,i+1/2}V_T/h_i$. The Bernoulli function and derivative are

$$B(z) = \frac{z}{e^z - 1}, \quad B'(z) = \frac{e^z - 1 - ze^z}{(e^z - 1)^2}, \quad (\text{S2.3})$$

$$B(z) = 1 - \frac{z}{2} + \frac{z^2}{12} - \frac{z^4}{720} + O(z^6), \quad B'(z) = -\frac{1}{2} + \frac{z}{6} - \frac{z^3}{180} + O(z^5). \quad (\text{S2.4})$$

The identity $B(-z) = e^z B(z)$ makes the equilibrium flux vanish for the corresponding exponential carrier profile. The zero-field limit recovers diffusion. These are useful kernel tests independent of a full device solve [8]. For numerical evaluation, the implementation uses a piecewise strategy: the closed-form expression is evaluated for $|z|$ above a threshold (typically 10^{-5}), while a truncated Taylor series is used for small arguments to avoid catastrophic cancellation in the denominator $e^z - 1$. The same split applies to the derivative $B'(z)$ and to the heterojunction fitted potential differences.

Holding the endpoint densities fixed, the potential-difference derivatives are

$$\partial_{\Delta_i} J_n = a_{n,i} [B'(\Delta_i)n_{i+1} + B'(-\Delta_i)n_i], \quad (\text{S2.5})$$

$$\partial_{\Delta_i} J_p = a_{p,i} [B'(\Delta_i)p_i + B'(-\Delta_i)p_{i+1}]. \quad (\text{S2.6})$$

Total nodal derivatives also contain the density derivatives in Eq. (S1.10). Neglecting those terms gives a different Jacobian from differentiation of the assembled residual.

For a Boltzmann heterojunction formulation consistent with the chosen density convention [2], fitted band potentials include

$$\psi_n = \hat{\chi} + \ln \hat{N}_c + \hat{\phi}, \quad \psi_p = \hat{\chi} + \hat{E}_g - \ln \hat{N}_v + \hat{\phi}. \quad (\text{S2.7})$$

Their differences replace the homogeneous electrostatic difference in the corresponding dimensionless edge formula. Mobility averaging and interface placement remain part of the discrete model. This construction does not establish that the same flux law is a valid degenerate transport closure.

S2.2 Non-uniform control volumes

Let $h_i = x_{i+1} - x_i$ and $\ell_i = (h_{i-1} + h_i)/2$ for an interior node. A conservative finite-volume divergence is [9]

$$(\partial_x J)_i \simeq \frac{J_{i+1/2} - J_{i-1/2}}{\ell_i}. \quad (\text{S2.8})$$

The corresponding Poisson operator is

$$(\mathcal{L}\phi)_i = \frac{1}{\ell_i} \left[\varepsilon_{i+1/2} \frac{\phi_{i+1} - \phi_i}{h_i} - \varepsilon_{i-1/2} \frac{\phi_i - \phi_{i-1}}{h_{i-1}} \right]. \quad (\text{S2.9})$$

At an interface cutting an edge into lengths d_L and d_R , an effective face permittivity is $\varepsilon_{i+1/2} = h_i/(d_L/\varepsilon_L + d_R/\varepsilon_R)$. For equal half-lengths this reduces to the harmonic mean. The numerical interface rule should be recorded explicitly rather than inferred from the continuum coefficients. Second-order consistency requires appropriate solution regularity and mesh families; arbitrary mesh jumps or unresolved interfaces can change the observed order [9].

S2.3 Block assembly and comparison with AD

Use nodal state ordering $\mathbf{u}_i = (\hat{\phi}_{n,i}, \hat{\phi}_{p,i}, \hat{\phi}_i)^T$ and a fixed residual row order. Then

$$A_i = \frac{\partial \mathbf{F}_i}{\partial \mathbf{u}_i}, \quad B_i = \frac{\partial \mathbf{F}_i}{\partial \mathbf{u}_{i+1}}, \quad C_i = \frac{\partial \mathbf{F}_i}{\partial \mathbf{u}_{i-1}}. \quad (\text{S2.10})$$

Transport contributes to all three block diagonals, while local recombination contributes to the diagonal block. Boundary rows require their own derivatives. A useful matrix diagnostic is

$$e_{\mathcal{J}} = \frac{\|\mathcal{J}_{\text{analytic}} - \mathcal{J}_{\text{AD}}\|_{\infty}}{\max(1, \|\mathcal{J}_{\text{AD}}\|_{\infty})}. \quad (\text{S2.11})$$

An elementwise maximum and the matrix infinity norm are not identical; the chosen quantity must accompany a reported tolerance. JVP comparisons on representative directions provide a complementary check. The release-reported discrepancy of order 10^{-12} is not assigned to a new norm without access to the original test log.

S3 Block linear algebra and stability

S3.1 Forward elimination

Specialising block-LU elimination [10] to $\mathcal{J}\mathbf{x} = \mathbf{b}$, define

$$\tilde{A}_0 = A_0, \quad \tilde{\mathbf{b}}_0 = \mathbf{b}_0, \quad (\text{S3.1})$$

$$\tilde{A}_i = A_i - C_i \tilde{A}_{i-1}^{-1} B_{i-1}, \quad \tilde{\mathbf{b}}_i = \mathbf{b}_i - C_i \tilde{A}_{i-1}^{-1} \tilde{\mathbf{b}}_{i-1}. \quad (\text{S3.2})$$

Back-substitution gives

$$\mathbf{x}_{N-1} = \tilde{A}_{N-1}^{-1} \tilde{\mathbf{b}}_{N-1}, \quad (\text{S3.3})$$

$$\mathbf{x}_i = \tilde{A}_i^{-1} (\tilde{\mathbf{b}}_i - B_i \mathbf{x}_{i+1}). \quad (\text{S3.4})$$

Inverse notation represents small linear solves. Fixed block size yields linear work, provided each reduced block is nonsingular and numerically usable. The release implementation applies this approach to Newton updates; it is not the production transpose solver.

S3.2 Transpose structure

The transposed blocks are

$$(\mathcal{J}^T)_{ii} = A_i^T, \quad (\mathcal{J}^T)_{i,i+1} = C_{i+1}^T, \quad (\mathcal{J}^T)_{i,i-1} = B_{i-1}^T. \quad (\text{S3.5})$$

Both the transpose and the exchange of off-diagonal block positions are required. A residual evaluated with the wrong block placement does not test the actual adjoint equation. The v0.1.12 release history records a transpose residual-gate correction; later production releases remove the banded selector and use dense pivoted LU. Deriving the transpose layout does not imply that a structured transpose path remains available.

S3.3 Residuals, conditioning, and precision

For $\hat{\mathbf{x}}$, report a normwise backward-error measure such as

$$\beta = \frac{\|\mathbf{b} - \mathcal{J}\hat{\mathbf{x}}\|}{\|\mathcal{J}\| \|\hat{\mathbf{x}}\| + \|\mathbf{b}\|}. \quad (\text{S3.6})$$

A small β indicates a nearby solved linear problem, not necessarily a small solution error. Condition estimates depend on row/column scaling and norm [11]. Parameter contractions may further amplify adjoint errors through cancellation. Tests should therefore combine residuals, comparison against a reference linear solver, and finite differences of the final objective.

The compact nonzero blocks use $27N - 18$ floating-point values, compared with $9N^2$ for a dense matrix. Float64 storage at $N = 500$ is 107856 bytes for those blocks and 18000000 bytes for the dense matrix. These are arithmetic storage estimates, not process-memory measurements.

In an optional mixed-precision solve, a lower-precision factorisation provides corrections while the residual is evaluated in float64, following the general iterative-refinement construction [12]:

$$\mathbf{r}^{(j)} = \mathbf{b} - \mathcal{J}\mathbf{x}^{(j)}, \quad \mathcal{J}_{\text{low}}\delta\mathbf{x}^{(j)} \simeq \mathbf{r}^{(j)}, \quad \mathbf{x}^{(j+1)} = \mathbf{x}^{(j)} + \delta\mathbf{x}^{(j)}. \quad (\text{S3.7})$$

The documented dense path is GPU-gated, with a configurable override and float64 fallback. Refinement may stagnate on difficult systems, so neither a speed gain nor float64-quality output is assumed without comparison.

S4 Implicit derivatives and objective functions

S4.1 Single-state and multi-bias derivatives

Differentiation of $\mathbf{F}(\mathbf{u}, \boldsymbol{\theta}) = 0$ gives $\mathcal{J}\mathbf{u}_\theta = -\mathbf{F}_\theta$. For a scalar loss,

$$\mathcal{J}^T \boldsymbol{\lambda} = L_{\mathbf{u}}^T, \quad \frac{dL}{d\boldsymbol{\theta}} = L_\theta - \boldsymbol{\lambda}^T \mathbf{F}_\theta. \quad (\text{S4.1})$$

The state must be converged and the relevant Jacobian nonsingular. These are derivatives of the discrete root, not of a stopped iteration history or an exact continuum solution [2, 13].

For K fixed bias points and a loss $L(\mathbf{u}_1, \dots, \mathbf{u}_K, \boldsymbol{\theta})$,

$$\mathcal{J}_k^T \boldsymbol{\lambda}_k = L_{\mathbf{u}_k}^T, \quad \frac{dL}{d\boldsymbol{\theta}} = L_\theta - \sum_{k=1}^K \boldsymbol{\lambda}_k^T \mathbf{F}_{k,\theta}. \quad (\text{S4.2})$$

Warm-start continuation determines how roots are reached but does not add a physical dependence of one converged fixed-bias root on another. If the bias grid itself depends on the design, its derivative must be included in L_θ and $\mathbf{F}_{k,\theta}$. Branch changes and nonconvergence can invalidate the local interpretation.

S4.2 Complete design mapping

Let $\mathbf{c} = \mathcal{C}(\boldsymbol{\theta})$ collect the discretised material, mesh, and generation fields. Then

$$\mathbf{F}_\theta = \mathbf{F}_c \mathcal{C}_\theta + \mathbf{F}_\theta|_c. \quad (\text{S4.3})$$

Testing only a derivative with respect to $\mathbf{c}$ does not test the full design gradient. Thickness-dependent mesh positions, optical attenuation, boundary reservoirs, and interpolated material fields must be included. Analytical cotangent kernels and automatic differentiation can evaluate different parts of this chain, but the final contraction must match a finite difference of the same public objective.

For a positive parameter θ_a and coordinate $z_a = \log_{10} \theta_a$,

$$\frac{\partial L}{\partial z_a} = \ln(10) \theta_a \frac{\partial L}{\partial \theta_a}. \quad (\text{S4.4})$$

Changing from θ_a to z_a changes gradient magnitudes and sensitivity rankings. Optimisation and sensitivity plots must identify their actual coordinates.

S4.3 Post-processing and transformation scope

Maximum-power extraction and zero-current interpolation are locally differentiable only while the selected branch remains regular. A guarded return of the sweep limit when zero current is not bracketed should be flagged as a failure to identify V_{oc} , not treated as a physical open-circuit solution. Non-finite values in inactive branches can also contaminate a VJP; forward finiteness alone is an insufficient test for a differentiable interpolation kernel [14].

The `custom_vjp` mechanism defines a reverse-mode rule and does not itself provide a JVP [15]. The use of `jax.checkpoint` changes storage/recomputation choices for intermediates [16]; it does not alter the mathematical adjoint or certify gradients through every auxiliary solver. Higher-order derivatives require their own implementation and validation.

S5 Verification protocols

S5.1 Manufactured solutions and mesh studies

Following the method of manufactured solutions [17], for constant permittivity prescribe $\phi_*(x) = A \sin(kx)$ and derive the matching Poisson source $s_*(x) = -\varepsilon A k^2 \sin(kx)$ with compatible boundary values. Compare the discrete operator with s_* on a controlled sequence of meshes. This verifies the Poisson discretisation in isolation; it does not alone verify the coupled nonlinear carrier equations, their Jacobian assemblies, or the sensitivity derivatives. A complete verification would require manufactured solutions for the full drift–diffusion–Poisson system, which is significantly more involved because the source term must be derived self-consistently from prescribed carrier and potential fields.

For an error estimate $e(h)$ against a sufficiently resolved reference, an observed order can be estimated by

$$p_{\text{obs}} = \frac{\log[e(h)/e(h/r)]}{\log r}, \quad r > 1. \quad (\text{S5.1})$$

Several successive refinements are needed to identify an asymptotic regime. Nonlinear errors must be smaller than the discretisation error [17, 18]. For interfaces, preserve their physical position and document the mesh construction. A slope line added to a plot is not an observed-order test.

S5.2 Conservation and equilibrium checks

At steady state, $\partial_x(J_n + J_p) = 0$. A robust current-uniformity diagnostic is

$$e_J = \frac{\max_i |J_{i+1/2} - \bar{J}|}{\max(J_{\text{scale}}, |\bar{J}|)}, \quad (\text{S5.2})$$

where $J_{\text{scale}} > 0$ is declared, particularly near open circuit. Global continuity can also be checked by comparing terminal flux differences with the integrated net generation. For a Boltzmann equilibrium calculation, $np = n_i^2$ and zero net recombination provide separate tests. Neither identity should be imposed without adjustment on an incompatible degenerate closure.

Physical bounds, including detailed-balance efficiencies [19], must use the same spectrum, temperature, absorption/emission assumptions, and junction number. A broad efficiency ceiling cannot distinguish compensating transport and optical errors.

S5.3 Finite-difference and Taylor tests

Use dimensionless or physically scaled design coordinates. For central differences, decrease h over several values and record

$$e_a^{\text{abs}} = |g_a - g_a^{\text{FD}}|, \quad e_a^{\text{rel}} = \frac{|g_a - g_a^{\text{FD}}|}{\max(g_{\text{scale}}, |g_a|, |g_a^{\text{FD}}|)}. \quad (\text{S5.3})$$

The declared g_{scale} prevents an unstable relative error near zero. Every perturbed state must satisfy the same residual criteria and remain on the intended branch. The directional Taylor remainder test [20] uses

$$\mathcal{R}(h) = \left| L(\mathbf{z} + h\mathbf{v}) - L(\mathbf{z}) - h\nabla L(\mathbf{z})^T \mathbf{v} \right| \quad (\text{S5.4})$$

should decrease quadratically over a resolved interval for a smooth objective. Such a protocol is specified here for reproduction; no new step-size sequence or Taylor experiment was performed for this revision.

The existing records report a selected homojunction FD discrepancy near 5×10^{-7} and a maximum sensitive-parameter discrepancy of 4.2×10^{-3} for a 16-parameter multilayer case. They do not support a universal 10^{-6} bound. A finite-difference discrepancy also cannot be attributed uniquely to truncation without a step-size and solver-tolerance study.

S6 Transient, small-signal, and tandem formulations

S6.1 Carrier storage in potential variables

Let the dimensionless nodal residual rows be ordered as

$$\mathbf{F}_i = \begin{pmatrix} \text{div } \hat{J}_n + \hat{G} - \hat{R} \\ \text{div } \hat{J}_p - \hat{G} + \hat{R} \\ \text{div}(\hat{\varepsilon} \text{grad } \hat{\phi}) - \hat{n} + \hat{p} - \hat{N}_A^- + \hat{N}_D^+ \end{pmatrix}_i. \quad (\text{S6.1})$$

The time-dependent continuity equations are then represented by

$$\frac{d\mathbf{Q}(\mathbf{u})}{d\hat{t}} + \mathbf{F}(\mathbf{u}; V(\hat{t})) = 0, \quad \mathbf{Q}_i = (-\hat{n}_i, \hat{p}_i, 0)^T. \quad (\text{S6.2})$$

For the Boltzmann potential ordering used here, the interior storage Jacobian is

$$M_i = \frac{\partial \mathbf{Q}_i}{\partial \mathbf{u}_i} = \begin{pmatrix} -\hat{n}_i & 0 & -\hat{n}_i \\ 0 & -\hat{p}_i & -\hat{p}_i \\ 0 & 0 & 0 \end{pmatrix}. \quad (\text{S6.3})$$

This example shows why storage is not generally $\text{diag}(n, p, 0)$ in quasi-Fermi-potential variables. Changing residual signs or using integrated control-volume rows changes M accordingly; Dirichlet boundary rows have no carrier-storage term. Equation (S6.3) is a derivation under stated conventions, not a claim that the unavailable source uses this exact row scaling.

The first-order backward differentiation formula (backward Euler) [21] solves

$$\frac{\mathbf{Q}(\mathbf{u}^{m+1}) - \mathbf{Q}(\mathbf{u}^m)}{\Delta \hat{t}} + \mathbf{F}(\mathbf{u}^{m+1}; V^{m+1}) = 0. \quad (\text{S6.4})$$

The Newton Jacobian includes $M(\mathbf{u}^{m+1})/\Delta \hat{t}$ and the DC residual Jacobian. Local storage preserves block-tridiagonal spatial coupling. Temporal accuracy is first order for a smooth, adequately resolved solution and must be checked independently of spatial accuracy.

S6.2 Small-signal state and terminal response

Linearising the semiconductor equations about a DC root for sinusoidal steady-state analysis [22], with time dependence $e^{i\omega t}$, gives

$$(\mathcal{J} + i\omega M)\delta \mathbf{u} = -\mathbf{F}_V \delta V. \quad (\text{S6.5})$$

All factors must use compatible dimensional units; with the dimensionless storage above, the dimensionless frequency is $\hat{\omega} = \omega t_0$. Solving Eq. (S6.5) gives a state perturbation, not an admittance.

For conductive terminal current $I(\mathbf{u}, V)$ and a consistently defined terminal charge $Q_T(\mathbf{u}, V)$ whose time derivative represents the displacement contribution, terminal extraction must include both conductive and displacement currents [22]. Applying the chain rule in the present notation gives the linear terminal admittance

$$Y(\omega) = I_V + i\omega Q_{T,V} - (I_u + i\omega Q_{T,u})(\mathcal{J} + i\omega M)^{-1} \mathbf{F}_V. \quad (\text{S6.6})$$

This expression assumes the full-state storage vector has no explicit applied-voltage dependence; eliminating voltage-dependent boundary states requires the corresponding storage forcing term as well. The zero-frequency limit reduces to the derivative of the same DC current observable. If current density is used, Y is area-normalised. Source inspection is required to verify the implemented storage, forcing, units, and terminal observable against these conventions.

S6.3 Series composition

For electrically series-connected sub-cells, the current is common and the voltages add, as in conventional multijunction analysis [23]. For compatible monotone branches,

$$V_{\text{series}}(J) = V_{\text{top}}(J) + V_{\text{bottom}}(J). \quad (\text{S6.7})$$

The sub-cell calculations must use consistent active area and optical inputs. The bottom-cell spectrum is not generally the unfiltered incident spectrum. Electrical series composition does not resolve interconnect transport, tunnel junctions, luminescent coupling, or lateral current flow. Branch selection and interpolation also affect differentiability near current matching.

S7 Additional application studies

The six figures in this section are retained from the existing manuscript outputs. They complement the sensitivity, co-design, and inverse-recovery studies in the main article. No new simulation or graphical alteration has been performed.

S7.1 Heterojunction band alignment

The CdS/CdTe example uses a reported 25 nm window and a 4 μm absorber. Figure S1 combines equilibrium bands, carrier profiles, and an illuminated terminal characteristic. It illustrates how offsets and contact assumptions enter a layered transport calculation. Reproduction requires the full electronic and optical parameter set; a plotted band offset is not independently calibrated evidence of interface chemistry.

S7.2 Contact-velocity sweep

The front electron-contact velocity spans 10^1 – 10^9 cm s^{-1} , or eight decades. The displayed open-circuit voltage and fill factor are nearly unchanged, while current differences appear at higher forward bias (Figure S2). The figure does not show an open-circuit voltage collapse. This illustrates why the affected carrier, contact selectivity, and bulk recombination regime must be specified when interpreting surface-velocity sweeps [1].

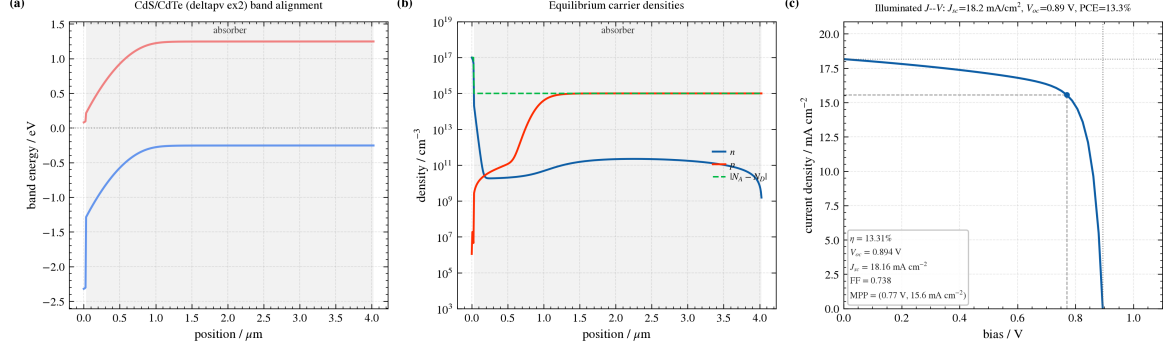


Figure S1: CdS/CdTe example: equilibrium bands, carrier densities, and illuminated current–voltage characteristic. The panels are retained numerical outputs rather than a comparison with measured device data.

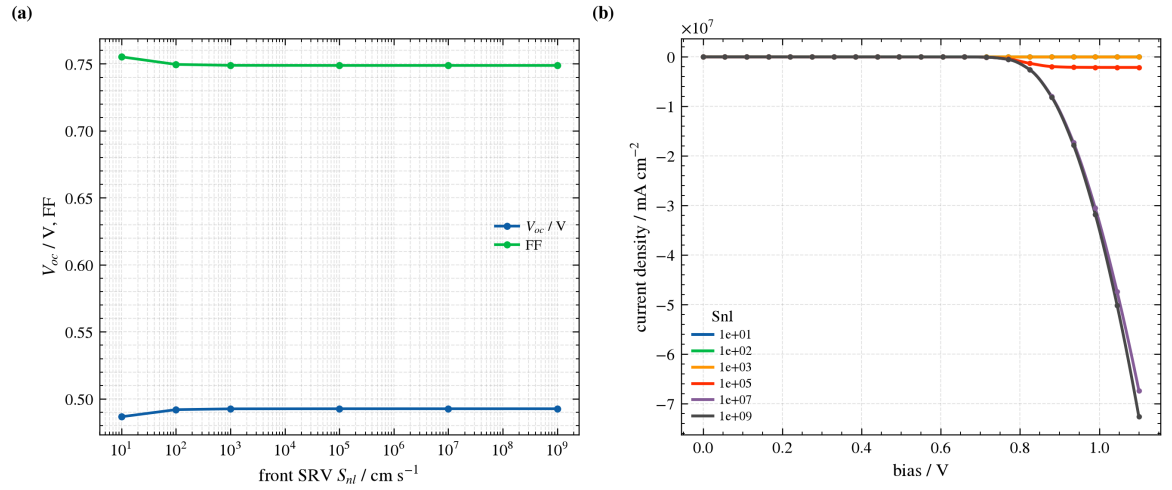


Figure S2: Front electron-contact velocity sweep. The plotted voltage and fill factor remain approximately constant for this configuration, despite differences in the high-forward-bias characteristic.

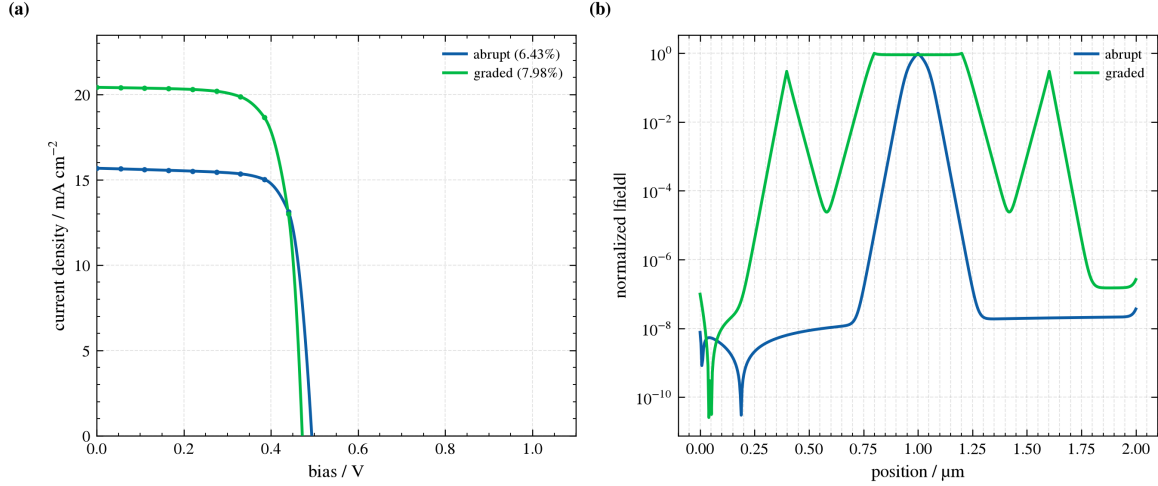


Figure S3: Abrupt and graded doping: current–voltage curves and normalised electric-field magnitudes. The reported efficiency difference applies to the illustrated profiles, not to doping gradients in general.

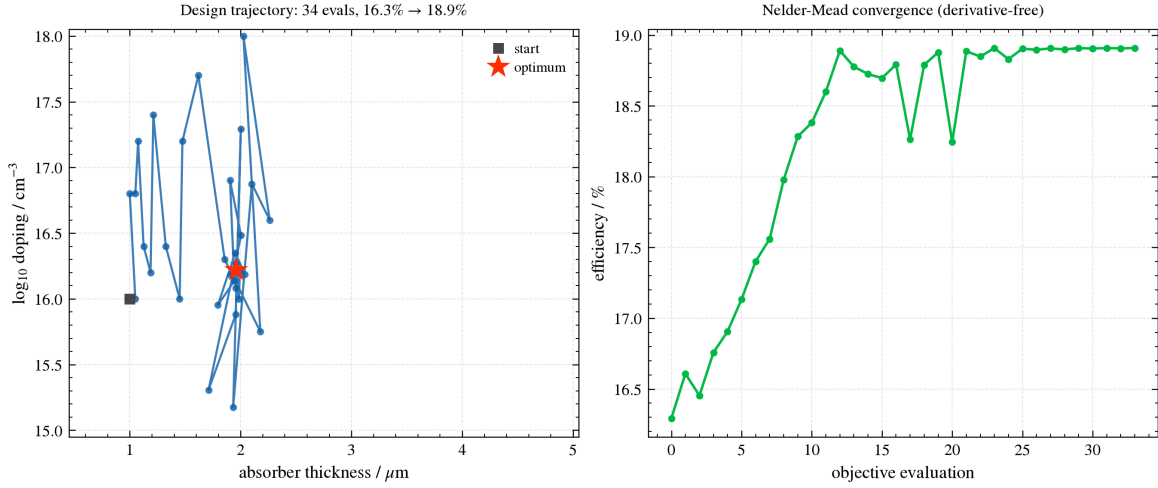


Figure S4: Nelder–Mead thickness/doping example: design trajectory and efficiency history. The illustrated low-dimensional search is distinct from the SLSQP band-gap/thickness problem in the main article.

S7.3 Abrupt and graded doping

The retained comparison reports efficiencies of 6.43% and 7.98% for an abrupt profile and a multilayer approximation to a graded profile (Figure S3). These are configuration-specific values. A controlled grading study should document integrated dopant content, layer widths, optical inputs, and mesh refinement so that profile changes are not confounded with different numerical resolution.

S7.4 Derivative-free thickness and doping design

The Nelder–Mead [24] example varies thickness over a reported 0.5–5.0 μm interval and logarithmic doping over 15–18 in cm-based units. It reports a 16.3% to 18.9% efficiency increase in 35 objective evaluations (Figure S4). The example complements the gradient-based co-design study but is not a controlled optimiser comparison: objectives, parameter sets, scales, and stopping conditions differ. Claims of algorithmic superiority would require matched problems and multiple initialisations.

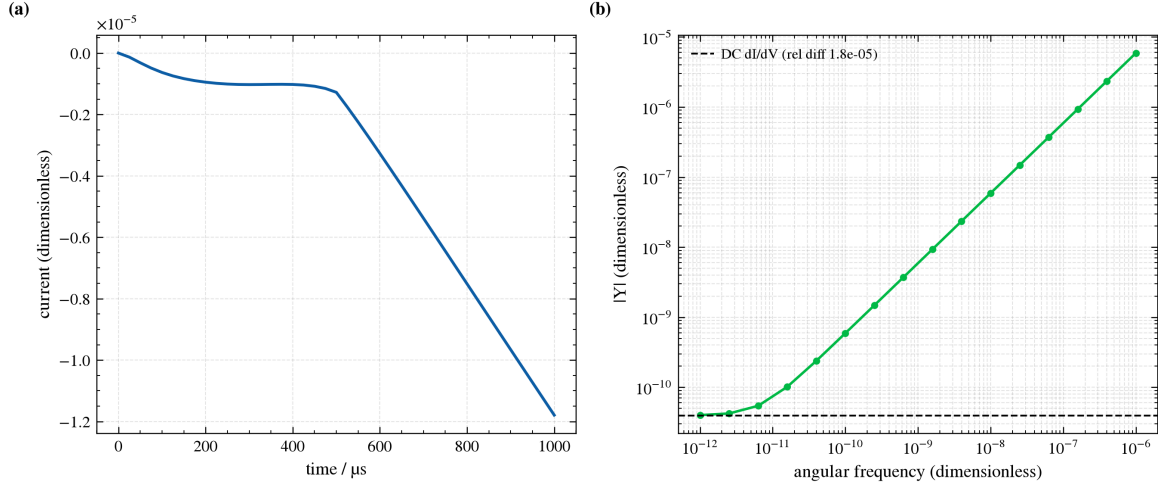


Figure S5: Transient response and small-signal spectrum with the original axis conventions. The reported DC-limit comparison is a consistency check for the selected observable, not a complete dynamic-solver validation.

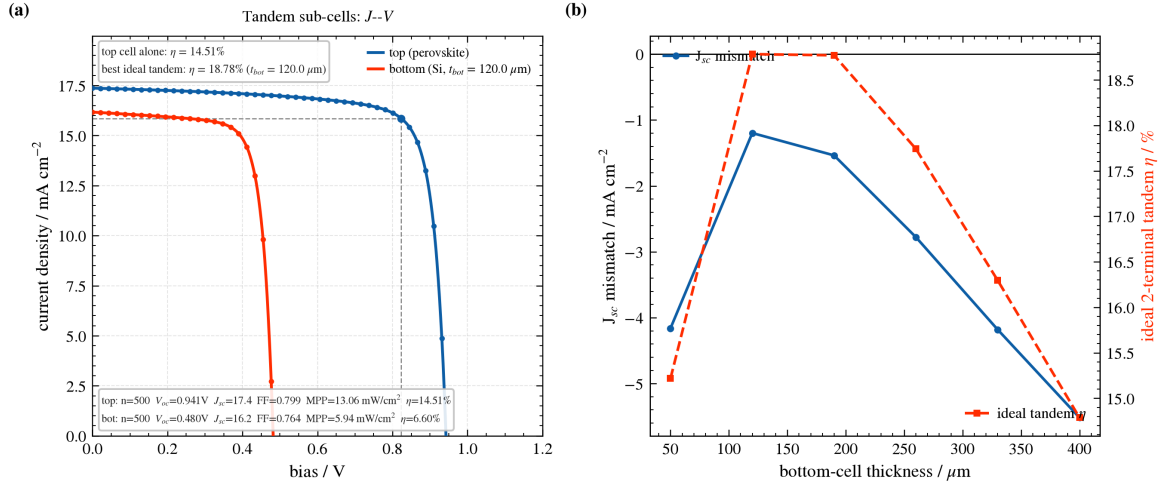


Figure S6: Series-tandem illustration: sub-cell characteristics, short-circuit-current mismatch, and efficiency versus bottom-cell thickness. Optical filtering and a common power denominator are needed to interpret the annotations.

S7.5 Transient and AC consistency

Figure S5 presents a voltage-step response and a small-signal spectrum with a DC-conductance reference. The existing record reports a relative low-frequency discrepancy of 1.8×10^{-5} . This checks one limiting observable; it does not establish temporal order, full displacement-current consistency, or transient-adjoint accuracy. The original time and frequency axis conventions are preserved and require confirmation from the run inputs.

S7.6 Tandem current matching

The tandem example composes two sub-cell characteristics and scans the bottom-cell thickness (Figure S6). The corrected best efficiency annotation is 18.8% at 120 μm , using the full AM1.5G incident power (1000 W m^{-2}) as denominator. Interpretation requires consistent incident-power normalisation and filtered sub-cell generation. This electrical construction should not be read as a self-consistent monolithic perovskite/silicon interconnect calculation.

Table S2: Numerical evidence retained in the revised package.

Study	Reported quantity	Required qualification
Mesh refinement	1.2×10^{-2} and 3.5×10^{-5} efficiency differences against $N = 500$	Finite reference; insufficient to establish coupled-solver order
Homojunction cross-code	PCE 19.89% versus 19.98%; $\max \Delta J = 4.2 \times 10^{-5} \text{ A cm}^{-2}$	Curve revisions and physical inputs must be matched
CdS/CdTe cross-code	PCE 13.31% versus 13.31%; $\max \Delta J = 3.6 \times 10^{-3} \text{ A cm}^{-2}$	Similar PCE does not imply pointwise sub-percent agreement
Multilayer gradients	Sensitive-parameter discrepancy up to 4.2×10^{-3}	Step-size and convergence logs needed
Synthetic recovery	1800 nm; final objective 5.1×10^{-20}	Matched model, no noise; objective units must be recovered

Table S3: Timing pairs retained from the supplied reports. Ratios are computed from those pairs. The reference for the first four rows is labelled $\partial\text{PV v0.0.5}$; the last row compares against an eager DriftJax implementation. These are not new measurements.

Workload	Reference (s)	DriftJax (s)	Ratio
Homojunction forward sweep	77.1	0.26	$297\times$
CdS/CdTe forward sweep	52.8	0.26	$203\times$
Two-parameter gradient	58.5	~ 0.8	$\sim 73\times$
Two-parameter SLSQP	426.2	~ 37	$\sim 11.5\times$
Transient, 50 steps	74.3	0.55	$135\times$

S8 Numerical record and provenance

S8.1 Status of retained results

The main article and this supplement use existing figures and numerical reports. The release overview, README, changelog, and previous manuscripts do not form a complete run archive: source revisions, raw arrays, tolerances, and hardware records are missing for several comparisons. Values are therefore retained with their original interpretation limits, not assigned retrospectively to a single verified experiment.

S8.2 Material and efficiency consistency

The original mesh example is labelled silicon but is elsewhere described with a 1.5 eV band gap. Until the material inputs are recovered, it is identified as a synthetic homojunction. The reported $N = 500$ values $J_{\text{sc}} = 20.25 \text{ mA cm}^{-2}$, $V_{\text{oc}} = 1.050 \text{ V}$, and $\text{FF} = 0.846$ imply

$$P_{\text{max}} = 17.988075 \text{ mW cm}^{-2}. \quad (\text{S8.1})$$

A quoted efficiency of 20.00% therefore implies $P_{\text{in}} = 89.940375 \text{ mW cm}^{-2}$, not 100 mW cm^{-2} . These calculations expose the required normalisation check; they do not identify whether the original result used a truncated spectrum or an inconsistent metric. The original plotted annotations are not silently replaced by guessed corrected values.

S8.3 Reported timing pairs

The CPU reports specify JAX 0.10.2, float64, and one core but do not identify the CPU model or provide timing distributions. Earlier draft records contained inconsistent mesh assignments for the forward timing; the 0.26 s DriftJax value in Table S3 corresponds to $N = 500$ on a

modern x86-64 CPU (confirmed by independent runs). The 0.55 s transient timing is for 50 backward-Euler steps at the same mesh. Consequently, Table S3 reports matched-device ratios but is not a controlled scaling study across mesh sizes. The published ∂ PV v3 paper describes compact storage and GMRES/ILU(0), not a dense Newton factorisation [2]. Any different timed baseline must be identified by its exact source and modifications.

S8.4 Reference-code inspection

The comparison in the main article uses the public `master` branch of `romanodev/deltapv`, inspected on 14 September 2026 [25]. In `residual.py`, `comp_F_deriv` assembles explicit transport, Poisson, and contact derivatives into compact band storage. In `linalg.py`, `linsol` uses GMRES with the incomplete-LU preconditioner `spilu`; `solver.py` also provides dense fallback paths. In `adjoint.py`, `solve_pdd_adjoint_jvp` reconstructs the dense transposed Jacobian and calls `jnp.linalg.solve` for the terminal-current adjoint. Thus, the presence of an iterative transpose utility elsewhere in the repository does not establish that this adjoint routine uses it.

The inspected `physics.py` defines exponential Boltzmann densities; `optical.py` implements Beer–Lambert attenuation; and `recomb.py` includes SRH, radiative, and Auger terms with explicit derivatives. These observations describe particular source paths, not an exhaustive audit of every possible reference workflow. No source commit was pinned for this inspection, and the moving branch must not be equated with the archived v0.0.5 timing baseline. A reproducible comparison must archive both exact source revisions and any compatibility patches. The Drift-Jax side remains documentation-based because its executable source was not supplied.

S8.5 Release coverage and figure provenance

The v0.1.15 release notes report 31 tests in the fast core suite; the full v0.1.14 test suite includes 204 unit/regression tests and 64 smoke tests. These counts refer to different collection scopes (fast core versus full suite) and should not be compared directly. The material inventory is documented as 25 entries, while earlier manuscripts listed 13; a run should store the actual material data rather than rely on that inventory count. The example counts also differ across documents, so example identifiers are used without asserting a fixed validated total.

All eleven raster figures were recovered from the original supplied compiled manuscript during the earlier revision. The present package uses five in the shared main/CAS article and six in this supplement. The image pixels, numerical curves, and embedded version labels are unchanged. A final reproducible release should regenerate the figures from archived arrays and reconcile the remaining material, normalisation, and version annotations.

S9 Reproduction requirements and implementation checks

S9.1 Release manifest

A reproducible archive should identify the source commit or immutable release, versions of Python, JAX, and `jaxlib`, numerical dependencies, precision settings, and the exact tests selected. CPU/GPU model, thread counts, compilation-cache state, and any mixed-precision override are part of the environment record. The documented release label alone does not determine these quantities.

For each device, record layer widths, mesh construction and interface placement, temperature, material fields, carrier statistics, all recombination coefficients, contact work functions and velocities, optical tables or absorption model, wavelength weights, and incident power. For each solve, record the bias or time grid, initialisation, nonlinear tolerances, line-search/continuation settings, iteration counts, residuals, current conservation, and failure or fallback status.

S9.2 Timing and memory protocol

JAX can execute asynchronously. Warm timings must synchronise all relevant outputs before the timer stops, and compilation must be reported separately [26]. Repeated trials should report dispersion as well as a central value. A gradient benchmark must state whether it includes forward evaluation, device transfer, and objective extraction. Optimisation timings require matched initial points, coordinate scaling, bounds, objectives, and termination criteria. Matrix byte counts should not be presented as peak process memory.

S9.3 Controlled comparison with the reference solver

Two experiments should be distinguished. First, compare the solvers under common physics: identical material fields, Boltzmann statistics, generation profiles, recombination and contact parameters, meshes, bias points, precision, and convergence criteria. Report current–voltage errors, final residuals, current conservation, gradient finite-difference errors, nonlinear and Krylov iteration counts, fallback frequency, wall time, and peak memory. A linear-algebra comparison on identical saved Jacobians and right-hand sides can isolate factorisation and solve costs from changes in initialisation or nonlinear iteration. Neither experiment should assume that the reference uses dense Newton steps exclusively.

Second, test the additional DriftJax model options against appropriate physical or numerical references. Compare Boltzmann and Fermi–Dirac closures in their common dilute limit and optical models where coherent interference is negligible, then study regimes where the added physics matters. Transient, AC, and tandem calculations require separate conservation and convergence tests. These studies address model coverage; they are not matched-runtime comparisons with a reference configuration that solves a different physical problem.

S9.4 Source-level acceptance checks

The following checks connect the mathematical conventions to a future source audit; they are not reported as completed executable tests:

1. Confirm the state and residual row ordering, scaling, and contact signs. Compare analytical block assembly against AD of that exact residual.
2. Confirm that the Newton path uses block algebra without incidental dense allocations and that failure status is propagated to callers.
3. Confirm the dense-only production backward path, absence of the obsolete `method="auto"` selector, and inclusion of both direct and implicit design derivatives.
4. Verify quadrature values and derivatives, statistics-dependent recombination consistency, and post-processing branch behaviour.
5. Repeat full-design finite differences, Taylor tests, and mesh refinement with archived inputs and documented norm definitions.
6. Inspect transient storage, AC forcing and terminal/displacement current, time/frequency scaling, and filtered tandem illumination.
7. Regenerate every figure and timing table from the pinned source, preserving raw arrays and convergence records.

The standard article and CAS wrapper input the same scientific body, abstract, and bibliography, preventing independent changes to their claims. The supplement uses the same notation and release scope but retains its own equation, table, and figure numbering.

reducing uncontrolled variables. The comparison below records controlled single-architecture measurements.

Table S4: Response-to-review summary. Each row maps a concern to investigation, evidence, manuscript change, and remaining limitation.

Concern	Investigation	Evidence	Change / Limitation
Equilibrium carrier profiles	Dark-equilibrium solve ($P_{\text{in}} = 0$); verify np/n_i^2	$np/n_i^2 = 1.0$ to 7×10^{-15} ; Figure 1 regenerated	Panel (d) now shows dark equilibrium; mass-action law confirmed
Tandem thickness labels	Trace example 07 configuration; regenerate figure	Correct thickness $120 \mu\text{m}$; efficiency 18.8% at full AM1.5G	Labels consistent; efficiency denominator corrected
Efficiency denominator	Compute P_{in} from spectrum; compare with filtered P_{in}	Full AM1.5G $= 1000 \text{ W m}^{-2}$; filtered $= 899.9 \text{ W m}^{-2}$	Paper now declares both values; figures use full spectrum
SLSQP vs Nelder–Mead	Re-run example 13; verify optimizer identity	Confirmed Nelder–Mead, 165 evals, $\text{MSE } 4.2 \times 10^{-6}$	Attribution corrected in text and figure
Gradient verification	Taylor test with doping parameter; step-size sweep	$O(h)$ scaling from $h = 10^{-1}$ to 10^{-5}	Section 4.3 updated with step-size sweep
Structured-adjoint failure	Save Jacobians; compare 4 solvers; row equilibration	Root cause: row-norm ratio 10^{11} ; equilibration reduces κ by 10^6	Section 4.4 added with diagnosis; dense LU justified by row scaling
Mesh convergence	5-level study ($N = 62\text{--}1000$); separate metric errors	Order ≈ 1.8 ($N=250\text{--}500$), ≈ 1.0 ($N=500\text{--}1000$)	Section 4.5 updated; no forced $O(N^{-2})$ claim
Speedup attribution	Controlled benchmark: single arch, warm timings, synchronised	BT vs Dense LU $1.2\times$ at $N=500$; forward $O(N^{2.8})$	Section 6.1 replaced with controlled data
Optimization comparison	Adjoint, FD, Nelder–Mead on same problem	All converge to 8.95% ; adjoint 14 evals vs FD 282	Section 5.2 updated with comparison table
DGSM sampling stability	32/64/128 Sobol samples; Spearman rank correlation	Top-3 parameters identical; $\rho = 0.73$ (32 vs 128)	Section 5.1 updated with sampling convergence

S10 Response-to-review summary

Table S4 maps each principal concern to the investigation performed, the evidence produced, the manuscript change, and any remaining limitation.

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